Eduardo **Alvarado**

Research Scientist in Human Motion and Avatar Synthesis - CV/ML/Graphics

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Education

Ph.D. in Human Motion - Computer Vision, Machine Learning & Graphics *Palaiseau, France*

École Polytechnique, LIX *Oct. 2020 - Oct. 2023*

- Thesis: Efficient Models for Human Locomotion and Interaction in Natural Environments.
- Supervisors: **Prof. Damien Rohmer** and **Prof. Marie-Paule Cani**.
- Funding: European Union's Horizon 2020 Research Programme. Marie Skłodowska-Curie Grant Agreement n. 860768 (CLIPE project).

M.Sc. in Embedded Systems and AI *Freiburg, Germany*

Albert-Ludwigs-Universität Freiburg *Apr. 2017 - Nov. 2019*

- Robotics, Computer Vision, Machine Learning, Deep Learning and Reinforcement Learning.
- Thesis: Deep Multi-modal Learning for Autonomous Driving.
- Supervisors: **Prof. Joschka Bödecker** and **Prof. Abhinav Valada**.

B.Sc. in Electronics and Automation Engineering *Madrid, Spain*

Universidad Carlos III - Poznan University of Technology (ERASMUS) *Sep. 2010 - Sep. 2014*

- Digital/Analog Electronics, Robotics and Control Engineering.
- Thesis: Localization-based AR Framework for Smart-glasses.
- Supervisors: **Prof. Angel Garcia Crespo**.

Academic & Industry Experience

Postdoctoral Researcher - Computer Vision for Virtual Avatars and Human Motion *Saarbrücken, Germany*

Max Planck Institute for Informatics *Jan. 2024 - Currently*

- Research on 3D/4D motion analysis, synthesis and 3D avatar generation. In-depth understanding of GenAI and multi-modal foundation models.
- Supervision of Ph.D. and M.Sc. students, leading research projects across institutions and defining technical directions.
- Advised by: **Prof. Christian Theobalt** and **Dr. Marc Habermann**.
- **Parental Leave Breaks**: 7 months (Sep. 2024 — Apr. 2025) and 5 months (Aug. 2026 — Dec. 2026).

Ph.D. Intern - Authoring Tools for Character Animation *Bordeaux, France*

Ubisoft La Forge *Mar. 2023 - Jun. 2023*

- Research on ML techniques for production-ready FK/IK algorithms for game development tools.

Research Scientist - AI for Material Rendering *Stuttgart, Germany*

InstaLOD GmbH *Jun. 2020 - Sep. 2020*

- Research on zero-shot AI methods for 3D material generation from single RGB images (smartphones).

M.Sc. Thesis - Deep Multi-modal Learning for Autonomous Driving *Renningen, Germany*

Robert Bosch GmbH *May 2019 - Nov. 2019*

- Research on sensor fusion for Autonomous Driving. Development of AI frameworks for Radar and RGB data acquisition and manipulation.

Research Assistant - Autonomous Intelligent Systems *Freiburg, Germany*

Albert-Ludwigs-Universität Freiburg *Sep. 2017 - Feb. 2019*

- TurtleBot3 Waffle hardware/software setups for teleoperation, SLAM and indoor mapping.

M.Sc. Intern - Dev-Tools Engineer for Autonomous Driving *Friedrichshafen, Germany*

ZF Friedrichshafen AG *Apr. 2018 - Oct. 2018*

- Development of computer vision ROS tools for vehicle data.

Business Strategy Analyst and QA Engineer for Robotics *Madrid, Spain*

BQ Mundo Reader S.L. *Nov. 2014 - May 2016*

- Responsible for Strategy Planning and Quality Monitoring. In-depth knowledge of Consumer Robotics. Lecturer for Robotics Workshops.

Languages

Spanish Native **English** Fluent **German** Fluent **French** Intermediate

Skills

Programming	Python, C#, C++, LaTeX	**Game/3D**	Unity, Unreal Engine, Blender	**Robotics**	ROS, Gazebo
AI	PyTorch, Tensorflow	**OS**	Windows, Linux	**SW-Dev**	CMake, Visual Studio, PyCharm, Git

Academic Activities

Research Visitor — *Barcelona, Spain*
UNIVERSITAT POLITÈCNICA DE CATALUNYA — *Sep. 2022 - Dec. 2022*
- Topic: Crowds Authoring in Natural Environments. Supervisor: Prof. Nuria Pelechano.

Research Visitor — *Nicosia, Cyprus*
CYENS CENTRE OF EXCELLENCE — *Jul. 2021 - Oct. 2021*
- Topic: Character Animation with focus on Motion Capture and Motion Matching. Supervisor: Prof. Yiorgos Chrysanthou.

Teaching Assistant for INF633 - Advanced 3D Graphics — *Palaiseau, France*
ÉCOLE POLYTECHNIQUE — *Oct. 2020 - Oct. 2023*
- Topic: Ecosystems Authoring, Procedural Animations, AI for Behavior Planning.

Reviewer / Committee — *Worldwide*
SIGGRAPH, SIGGRAPH ASIA, IEEE VR, TVCG, EUROGRAPHICS (PC), CEIG (PC), CASA (PC) — *Oct. 2020 - Currently*

Selected Publications

PEER-REVIEWED JOURNAL AND CONFERENCE ARTICLES

SOMA: From Surface Observations to Muscle Anatomy ()
EDUARDO ALVARADO, EMILY KIM, GERRIT NOLTE, FRIEDEMANN RUNTE, MARIO BOTSCH, MARC HABERMANN, CHRISTIAN THEOBALT
European Conference on Computer Vision (ECCV), 2026

SoleSense: Insole-guided Stability for Diffusion-driven Humanoids
SIDHARTH SURESH NAIR, **EDUARDO ALVARADO**, CHRISTIAN THEOBALT
(Under Review), 2026

Towards Real-World Wearable Motion Reconstruction ()
ANDREA BOSCOLO CAMILETTO, RISHABH DABRAL, **EDUARDO ALVARADO**, THABO BEELER, MARC HABERMANN, CHRISTIAN THEOBALT
European Conference on Computer Vision (ECCV), 2026

Step2Motion: Locomotion Reconstruction from Pressure Sensing Insoles ()
JOSE LUIS PONTON, **EDUARDO ALVARADO**, LIN GENG FOO, NURIA PELECHANO, CARLOS ANDUJAR, MARC HABERMANN
Eurographics, 2026

FRAME: Floor-aligned Representation for Avatar Motion from Egocentric Video ()
ANDREA BOSCOLO CAMILETTO, JIAN WANG, **EDUARDO ALVARADO**, RISHABH DABRAL, THABO BEELER, MARC HABERMANN, CHRISTIAN THEOBALT
Computer Vision and Pattern Recognition (CVPR), 2025

BimArt: A Unified Approach for the Synthesis of 3D Bimanual Interaction with Articulated Objects ()
WANYUE ZHANG, RISHABH DABRAL, VLADISLAV GOLYANIK, VASILEIOS CHOUTAS, **EDUARDO ALVARADO**, THABO BEELER, MARC HABERMANN, CHRISTIAN THEOBALT
Computer Vision and Pattern Recognition (CVPR), 2025

TRAIL: Simulating the Impact of Human Locomotion on Natural Landscapes ()
EDUARDO ALVARADO, OSCAR ARGUDO, DAMIEN ROHMER, MARIE-PAULE CANI, NURIA PELECHANO
Computer Graphics International (CGI), 2024

Generating Upper-Body Motion for Real-Time Characters Making their Way through Dynamic Environments ()
EDUARDO ALVARADO, DAMIEN ROHMER, MARIE-PAULE CANI
Proceedings of the ACM SIGGRAPH/Eurographics Symposium on Computer Animation (SCA), 2022 — ***Best Paper Honorable Mention Award***

Real-Time Locomotion on Soft Grounds With Dynamic Footprints ()
EDUARDO ALVARADO, CHLOÉ PALIARD, DAMIEN ROHMER, MARIE-PAULE CANI
Frontiers in Virtual Reality, Frontiers, 2022, 3, (10.3389/frvir.2022.801856)

A Survey on Reinforcement Learning Methods in Character Animation ()
ARIEL KWIATKOWSKI, **EDUARDO ALVARADO**, VICKY KALOGEITON, KAREN C. LIU, JULIEN PETTRÉ, MICHIEL VAN DE PANNE, MARIE-PAULE CANI
Eurographics 2022, Computer Graphics Forum, Wiley, 2022, pp.1-27. (10.1111/cgf.14504)

Interests

I'm passionate about skiing and try to explore a new destination each year with friends and family. I stay actively engaged with the latest developments in robotics and AI, often sharing my curiosity with my son to spark his interest. In my spare time, I work on building my own robotics startup, alongside learning piano, doing DIY projects, and exploring game-dev.